

**Name and Strength of Active Substance(s):**

Each APOQUEL 3.6mg tablet contains 3.6mg of oclacitinib (as oclacitinib maleate).

Each APOQUEL 5.4mg tablet contains 5.4mg of oclacitinib (as oclacitinib maleate).

Each APOQUEL 16mg tablet contains 16mg of oclacitinib (as oclacitinib maleate).

Product Description:

APOQUEL 3.6mg tablet: White to off-white caplet shaped with “AQ”, single score line and “S” on both sides.

APOQUEL 5.4mg tablet: White to off-white caplet shaped with “AQ”, single score line and “M” on both sides.

APOQUEL 16mg tablet: White to off-white caplet shaped with “AQ”, single score line and “L” on both sides.

Pharmacodynamics:

Oclacitinib is a Janus kinase (JAK) inhibitor. It can inhibit the function of a variety of cytokines dependent on JAK enzyme activity. For oclacitinib, the target cytokines are those that are proinflammatory or have a role in allergic responses/pruritis. However, oclacitinib may also exert effects on other cytokines (for example, those involved in host defence or haematopoiesis) with the potential for unwanted effects.

Pharmacokinetics:

Following oral administration in dogs, oclacitinib maleate is rapidly and well absorbed, with a time to peak plasma concentration (t_{max}) of less than 1 hour. The absolute bioavailability of oclacitinib maleate was 89%. The prandial state of the dog does not significantly affect the rate or extent of its absorption.

Total body oclacitinib clearance from plasma was low – 316 ml/h/kg bodyweight (5.3 ml/min/kg bodyweight), and the apparent volume of distribution at steady-state was 942 ml/kg bodyweight.

Following intravenous and oral administration, the terminal $t_{1/2}$ s were similar at 3.5 and 4.1 hours respectively. Oclacitinib exhibits low protein binding with 66.3% to 69.7% bound in fortified canine plasma at nominal concentrations ranging from 10 to 1,000 ng/ml.

Oclacitinib is metabolised in the dog to multiple metabolites. One major oxidative metabolite was identified in plasma and urine.

Overall the major clearance route is metabolism, with minor contributions from renal and biliary elimination. Inhibition of canine cytochrome P450s is minimal with IC_{50} s 50-fold greater than the observed mean C_{max} (333 ng/ml or 0.997 μ M) following 0.6 mg/kg bw oral administration in the target animal safety study. Therefore, the risk of metabolic drug-drug interactions due to oclacitinib inhibition is very low. No accumulation was observed in the blood of dogs treated for 6 months with oclacitinib.

Indications:

Control of pruritus associated with allergic dermatitis and control of atopic dermatitis in dogs at least 12 months of age.

Target Species:

Dogs

Recommended Dose:

The dose of APOQUEL tablets is 0.4 to 0.6 mg oclacitinib/kg body weight, administered orally, twice daily for up to 14 days, and then administered once daily for maintenance therapy. APOQUEL may be administered with or without food.

Dosing Chart:

Weight Range (in Kg)		Number of Tablets to be Administered		
Low	High	3.6mg Tablets	5.4mg Tablets	16mg Tablets
3.0	4.4	0.5	-	-
4.5	5.9	-	0.5	-
6.0	8.9	1	-	-
9.0	13.4	-	1	-
13.5	19.9	-	-	0.5
20.0	26.9	-	2	-
27.0	39.9	-	-	1
40.0	54.9	-	-	1.5
55.0	80.0	-	-	2

Contraindications:

Do not use in cases of hypersensitivity to the active substance or to any of the excipients.

Do not use in dogs less than 12 months of age or less than 3 kg bodyweight.

Do not use in dogs with evidence of immune suppression, such as hyperadrenocorticism, or with evidence of progressive malignant neoplasia as the active substance has not been evaluated in these cases.

Warnings & Precautions:**Warnings:**

APOQUEL is not for use in dogs less than 12 months of age.

APOQUEL is not for use in dogs with serious infections.

APOQUEL may increase susceptibility to infection, including demodicosis, and exacerbate neoplastic conditions (see Adverse Reactions).

Human Warnings:

Not for human use.

Keep this and all drugs out of reach of children.

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

- Wash hands after administration.
- In case of accidental ingestion, seek medical advice immediately.

Precautions:

APOQUEL is not for use in breeding dogs, or pregnant or lactating bitches.

The use of APOQUEL has not been evaluated in combination with glucocorticoids, cyclosporine, or other systemic immunosuppressive agents. Dogs receiving APOQUEL should be monitored for the development of infections, including demodicosis, and neoplasia.

For Animal Use Only.
Controlled Medicine / Ubat Terkawal
Keep out of reach of children / Jauhi dari kanak-kanak.

Interactions with Other Medicaments:

No drug interactions were observed in field studies where oclacitinib was administered concomitantly with veterinary medicinal products such as endo- and ectoparasiticides, antimicrobials and antiinflammatories.

The impact of oclacitinib administration on vaccination with modified live vaccines, canine parvovirus (CPV), canine distemper virus (CDV) and canine parainfluenza (CPI) and inactivated rabies vaccine (RV), on 16 week old vaccine naïve puppies has been studied. An adequate immune response (serology) to CDV and CPV vaccination was achieved when puppies were administered oclacitinib at 1.8 mg/kg bodyweight twice daily for 84 days. However, the findings of this study indicated a reduction in serological response to vaccination with CPI and RV in puppies being treated with oclacitinib compared to untreated controls. The clinical relevance of these observed effects for animals vaccinated while being administered oclacitinib (in accordance with the recommended dosing regimen) is unclear.

Pregnancy and Lactation:

The safety of the veterinary medicinal product has not been established during pregnancy and lactation, or in breeding male dogs, therefore its use is not recommended during pregnancy, lactation or in dogs intended for breeding.

Adverse Reactions / Side Effects:

The common adverse reactions seen up to day 16 of the field trials are listed in the following table:

	Adverse reactions observed in atopic dermatitis study up to Day 16		Adverse reactions observed in pruritus study up to Day 7	
	Apoquel (n=152)	Placebo (n=147)	Apoquel (n=216)	Placebo (n=220)
Diarrhoea	4.6%	3.4%	2.3%	0.9%
Vomiting	3.9%	4.1%	2.3%	1.8%
Anorexia	2.6%	0%	1.4%	0%
New cutaneous or subcutaneous lumps	2.6%	2.7%	1.0%	0%
Lethargy	2.0%	1.4%	1.8%	1.4%
Polydipsia	0.7%	1.4%	1.4%	0%

After day 16, the following adverse reactions have been observed:

- pyoderma and non-specified dermal lumps have been observed very commonly;
- otitis, vomiting, diarrhoea, histiocytoma, cystitis, yeast skin infections, pododermatitis, lipoma, polydipsia, lymphadenopathy, nausea, increased appetite and aggression have been observed commonly.

Treatment-related clinical pathology changes were restricted to an increase in mean serum cholesterol and a decrease in mean leukocyte count, however, all mean values remained within the laboratory reference range. The

decrease in mean leukocyte count observed in oclacitinib-treated dogs was not progressive, and affected all white blood cell counts (neutrophil, eosinophil and monocyte counts) except lymphocyte counts. Neither of these clinical pathology changes appeared clinically significant. The development of papillomas was noted in a number of dogs in a laboratory study. Anaemia and lymphoma have been reported very rarely in spontaneous reports.

Regarding susceptibility to infection and neoplastic conditions, see section *Precautions for Use*.

The frequency of adverse reactions is defined using the following convention:

- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

Overdose:

Oclacitinib tablets were administered to healthy, one year old Beagle dogs twice daily for 6 weeks, followed by once per day for 20 weeks, at 0.6 mg/kg bw, 1.8 mg/kg bw and 3.0 mg/kg bw for a total of 26 weeks.

Clinical observations that were considered likely to be related to oclacitinib treatment included: alopecia (local), papilloma, dermatitis, erythema, abrasions and scabbing/crusts, interdigital "cysts", and oedema of the feet.

Dermatitis lesions were mostly secondary to the development of interdigital furunculosis on one or more feet during the study, with the number and frequency of observations increasing with increasing dose. Lymphadenopathy of peripheral nodes was noted in all groups, increasing in frequency with increasing dose, and was frequently associated with interdigital furunculosis.

Papilloma was considered treatment related, but not dose related.

There is no specific antidote and in case of signs of overdose the dog should be treated symptomatically.

Storage Condition: Store below 30°C.

Shelf Life (as packaged for sale): 3 years.

Shelf-life after first opening the container:

Proposed Shelf Life of remaining ½ tablet in the HDPE bottle:

Consume the remaining ½ tablet within 14 days of breaking.

Pack Size: Bottle of 100's count.

Disposal: The material should be disposed in compliance with national and regional requirements.

Manufactured by:

Zoetis Inc

2605 East Kilgore Road, Kalamazoo, Michigan 49001, United States

Product Registration Holder:

Zoetis Malaysia Sdn bhd

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